Accelerating Deep Learning Algorithms on Memristors

Addressing Reliability Issues and Hardware-aware Pruning

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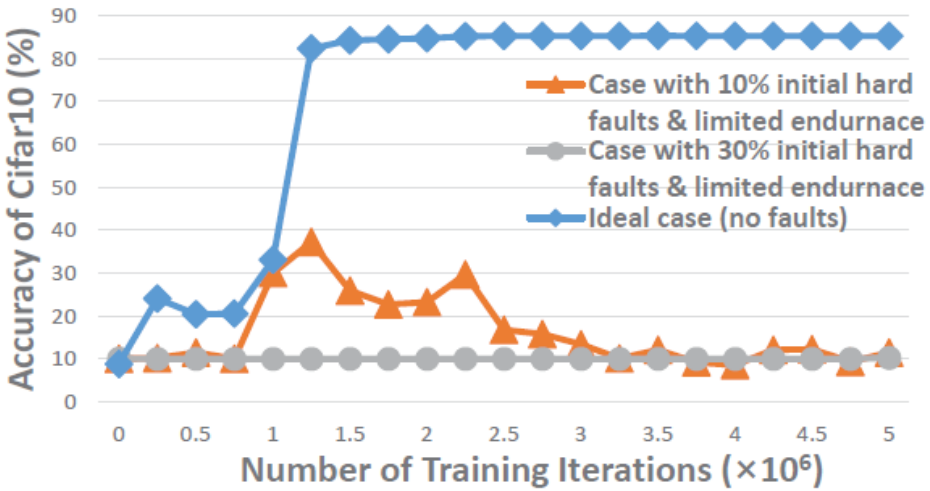
Background on reliability issues

- **Process variation:** deviation in parameters from their nominal values
- **Hard error:** a cell is stuck at the value 0 or 1, which happens when the write endurance limit of a cell has been reached
- **Resistance drift:** change in the resistance of the cell over time which leads to a soft-error

Reliability issues

- High defect rate & process variation =>reliability issues
- Especially for large neuromorphic systems, a ReRAM implementation leads to heavy wire congestion and poor reliability of read/write operations due to voltage-drop
- With increasing ReRAM failure rate, accuracy of NN reduces drastically



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Challenges in achieving high accuracy & perf.

- Compared to SRAM, ReRAM has high write energy & latency which increases overall power consumption
- During NN training, precise tuning of ReRAM requires frequent update of weights and large number of training iterations for convergence
- This leads to high number of writes and large energy consumption
- Increasing number of training iterations leads to energy penalty and aggravates ReRAM endurance issues
- Retraining may not be sufficient in case of high fault rate

Techniques for Mitigating Hard faults (stuck-at-0 or stuck-at-1)

Using NN retraining (1 of 2)

- Challenge: large weight may get mapped to a ReRAM with high variation
- **Observations**: In NNs, loss in accuracy due to removal of some weights is automatically compensated during training by changing the neighboring weights (called self-healing capability)

- Technique: Train NN based on distribution of stuck-at-faults and resistance variation in crossbar.
- Lower the magnitude of weight mapped to a memristor with large variation/fault-rate

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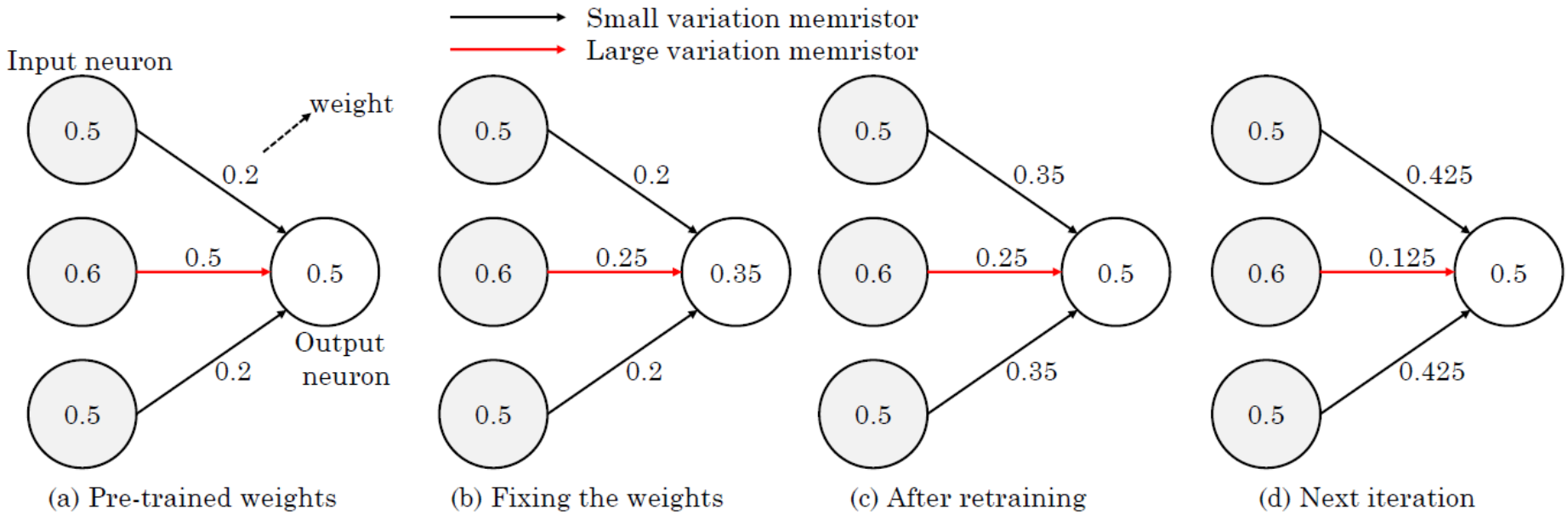
Using NN retraining (2 of 2)



This technique can remove nearly all the loss in NN accuracy due to faults/variations in ReRAM

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Thresholding and Remapping

- **Observation 1:** Small weights make little contribution to the accuracy, but reduce ReRAM lifetime significantly

- **Idea 1:** In back-propagation, flush weights smaller than a threshold to zero
- This strategy boosts ReRAM lifetime while bringing only small increase in the number of training iterations

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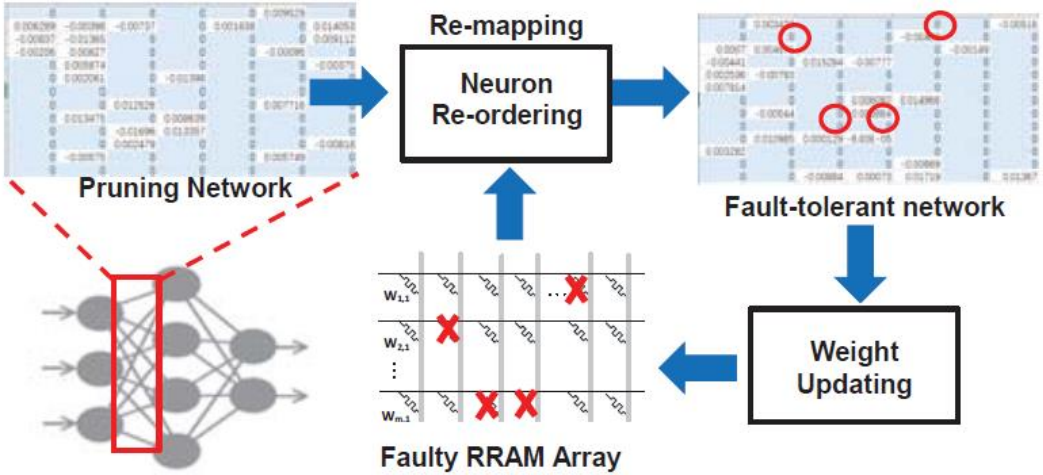
Thresholding and Remapping

- **Observation 2:** Pruning leads to a large fraction of weights becoming zero
- **Idea 2:** Use this to tolerate stuck-at-0 faults.
- For this, reorder rows/columns of the weight matrix so as to map zero weights to cells having stuck-at-0 faults



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Weight significance based training

- Classify the weights into crucial and non-crucial based on their impact on final accuracy
- Perform retraining for recovering the accuracy loss due to faults: Identify stuck weights and retune remaining weights for recovering accuracy.
- Instead of giving random values to weight matrices, use pre-trained weight matrix for initializing and accelerating re-training.
- If error rate too high, use redundant columns. Remap only crucial weights to them.
- Result: By remapping only few weights, large improvement in accuracy is achieved.

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Techniques for Mitigating Resistance Drift

Adaptive tuning of memristor (1 of 2)

- Challenge: Memristor gradually drifts from its original programmed state,
- e.g., a voltage of 0.1 V leads to 2% deviation from the original state in 1 s.
- => Precision of an MCA-based system decreases gradually
- Idea: Tune memristor to a specific state to compensate this effect
- Challenge: Memristor tuning takes much more latency than a regular operation (5 ms vs. 50 ns)
- => frequent calibration incurs large overhead

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Adaptive tuning of memristor (2 of 2)

- Idea: Observe error values over time.
- Fit an Nth-degree polynomial to show relationship between time and computation error
- Predict error value at next time instance.
- If this is less than the maximum tolerable error, skip calibration.
- Otherwise, predict the time instance at which the predicted error would have reached maximum tolerable error. Then, perform calibration at this time instance
- Result: tuning requirement reduced.

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MCA-configuration aware pruning
and
using suitable MCA-configuration for a pruned network

MCA-aware pruning strategy (1 of 2)

- Challenge: Pruning techniques lead to a network with irregular connections, which may not match to MCA design
- => inefficient hardware implementation

Input	Connectivity map			
I1	1	1	1	1
I2	1	1	1	1
I3	1	1	1	1
I4	1	1	1	1
Output	O1	O2	O3	O4

Original network

0	1	0	1
1	0	1	0
0	1	0	1
1	0	1	0

Pruned network

Here, 1/0 shows presence/absence of connection and not the actual weight

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MCA-aware pruning strategy (2 of 2)

- Idea: Prune at the granularity of MCA and not weights

1	1	0	0
1	1	0	0
0	0	1	1
0	0	1	1

- Helps in retaining the gains of algorithmic pruning and network sparsity in memristor-based NN-engines
- Their technique is applicable to FC layers, which show no data reuse and have most of the weights of the network

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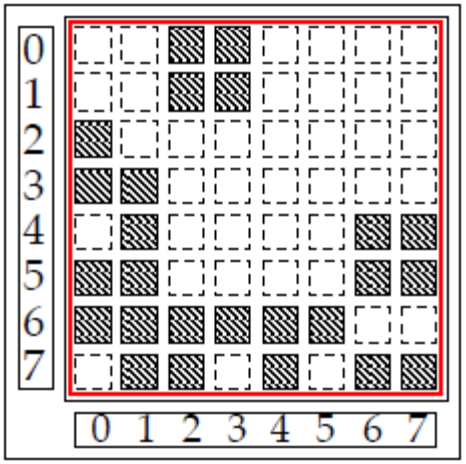
Impact of MCA size (1 of 2)

- MCA size decides tradeoff between parallelism/speedup/cost and utilization
- Consider an 8X8 adjacency matrix.
- Shadowed square = an edge between two vertices
- Blank square = no edge between two vertices

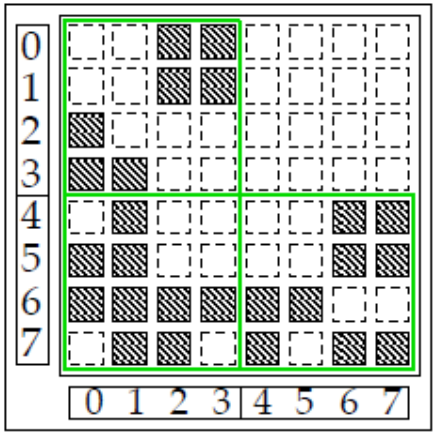


(a) 8-8 Mapping

On using 8*8 MCA, utilization is only 39.1% (=25/64)

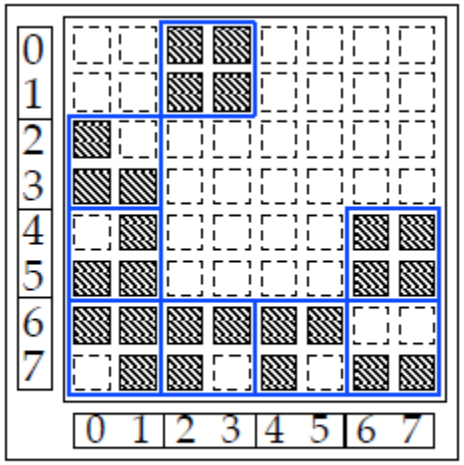
Impact of MCA size (2 of 2)

- With 4*4 MCA, utilization is 52.1% (=25/48) since subgraph in upper-right corner need not be mapped to MCA.



(b) 4-4 Mapping

- With 2*2 MCA, utilization is increased to 78.1% (=25/32)



(c) 2-2 Mapping

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Solved question

- In a 6*6 adjacency matrix, following weights are zero : (0, 1), (0,5), (1, 3), (4,4), (2,0), (5,5), (4,1), (4,5), (5,2), (5,4).
- Note that index is shown as (x,y), where x and y each go from 0 to 5.

	zero				zero
			zero		
zero					
	zero			zero	zero
		zero		zero	zero

Utilization on using 3*3 MCA is

	zero				zero
			zero		
zero					
	zero			zero	zero
		zero		zero	zero

Answer: 26/36

Utilization on using 2*2 MCA is

	zero				zero
			zero		
zero					
	zero			zero	zero
		zero		zero	zero

Answer: 26/32

References

- S. Mittal, “A Survey of ReRAM-based Architectures for Processing-in-memory and Neural Networks”, Machine learning and knowledge extraction, 2018 (pdf)